



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

Full Syllabus

23-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	C	C	D	B	B	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	C	B	B	C	C	D	B	A	D
	Q.	1	2	3	4	5	6	7	8	9	10
A.	9.60	23.60	1.50	300.00	11.09 to 11.10	7.00	34.80	3.75	4.50	0.10	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	D	D	C	A	C	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	A	C	D	D	B	A	A	C
	Q.	1	2	3	4	5	6	7	8	9	10
A.	16.00	16.00	160.00	400.00	1.24	3.00	11.00	1.00	0.00	2.00	

PART-3 : MATHEMATICS

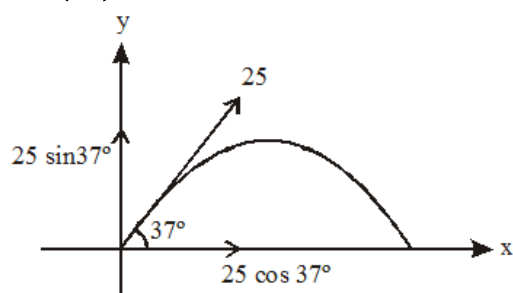
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	B	C	B	C	D	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	C	D	B	D	C	D	B	A
	Q.	1	2	3	4	5	6	7	8	9	10
A.	2.00	4.25	1.00	7.50	1.50	5.33	2.50	0.00	1.00	0.00	

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)



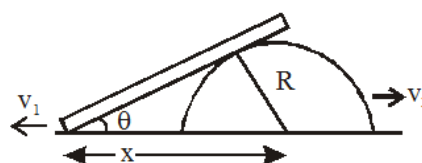
$$x = 25 \times \frac{4}{5} \times 2 = 40 \text{ m}$$

$$u_y = 25 \times \frac{3}{5} = 15$$

$$y = 15 \times 2 - \frac{1}{2} 10 (4) = 10 \text{ m}$$

$$\vec{r} = 40\hat{i} + 10\hat{j}$$

2. Ans (A)



$$\sin \theta = \frac{R}{x}$$

$$x = R \sec \theta$$

$$\frac{dx}{dt} = -R \sec \theta \cot \theta \frac{d\theta}{dt}$$

$$\frac{dx}{dt} = (v_1 + v_2)$$

$$\therefore (v_1 + v_2) = R \sec \theta \cot \theta \frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = \left(\frac{(v_1 + v_2) \sin^2 \theta}{R \cos \theta} \right)$$

3. **Ans (D)**

$$m_1 = m_n$$

$$m_2 = Am_n$$

$$v_1 = v \quad v_2 = 0$$

$$v_1 = \frac{(m_n - Am_n)}{m_n + Am_n} v + \frac{2Am_n}{m_n + Am_n} \times 0$$

$$v_1 = \left(\frac{1 - A}{1 + A} \right) v$$

$$v_f = \frac{1}{2} m_n v_1^2$$

$$v_i = \frac{1}{2} m_n v^2$$

$$\frac{v_f}{v_i} = \left(\frac{1 - A}{1 + A} \right)^2$$

4. **Ans (C)**

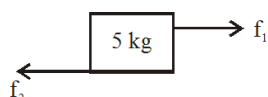
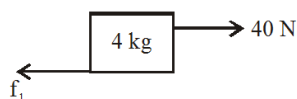
$$v = \sqrt{Rg \tan \theta}$$

$$80 \times \frac{5}{18} = \sqrt{200 \times 10 \tan \theta}$$

$$\tan \theta = \frac{20}{81}$$

$$\theta = \tan^{-1} \left(\frac{20}{81} \right)$$

6. **Ans (D)**



$$40 - 27 = 9a$$

$$a = \frac{13}{9} \text{ m/sec}^2$$

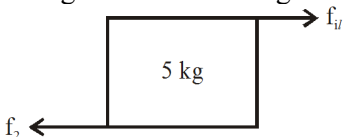
$$40 - f_1 = 4 \left(\frac{13}{9} \right)$$

$$f_1 = 34.22 \text{ N but } f_{1\ell} = 20 \text{ N}$$

(Limiting friction force)

$$\therefore f_1 > f_{1\ell}$$

$\therefore$ 4 kg will slide on 5 kg.



$$20 - 27 = -ve$$

$$\therefore f_{1\ell} < f_2$$

$\therefore$ 5 kg will not slide or move

$\therefore$ 4 kg will move with acceleration

$$\frac{40 - 20}{4} = 5 \text{ m/sec}^2$$

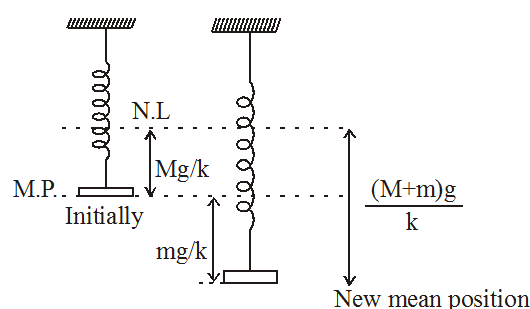
and 5 kg will not move.

7. **Ans (B)**

By momentum conservation

$$mv = (M + m)v'$$

$$v' = \frac{mv}{M + m}$$



$$v'^2 = \omega^2 (A^2 - x^2)$$

$$x = \frac{mg}{k}$$

$$\omega^2 = \frac{k}{M + m}$$

$$v'^2 = \omega^2 (A^2 - x^2)$$

$$\frac{m^2 v_0^2}{(M + m)^2} = \frac{k}{M + m} \left(A^2 - \frac{m^2 g^2}{k^2} \right)$$

$$A^2 = \frac{m^2 v_0^2}{k(M + m)} + \frac{m^2 g^2}{k^2}$$

$$A = \frac{m}{k} \sqrt{\frac{v_0^2 k}{M + m} + g^2}$$

8. **Ans (B)**

$$v_P = \sqrt{2g \times h}$$

$$(\text{Range})_P = v_P \times t = v_P = \sqrt{2gh} \times \sqrt{\frac{2 \times 3h}{g}}$$

$$= 2\sqrt{3}h \dots (i)$$

Bernoulli's theorem between surface and Q

$$2\rho g \times 2h + 4\rho g \times 2h = \frac{1}{2} 4\rho v_Q^2 + 4\rho gh$$

$$v_Q = \sqrt{4gh} = 2\sqrt{gh}$$

$$(\text{Range})_Q = \sqrt{4gh} \times \sqrt{\frac{2 \times h}{g}}$$

$$\Rightarrow 2\sqrt{2}h$$

$$\frac{R_P}{R_Q} = \frac{2\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}}$$

9. **Ans (A)**

$$x = 15 \sin \omega t - 20 \sin^3 \omega t$$

$$= 15 \sin \omega t - 5 (4 \sin^3 \omega t)$$

$$(\because 4 \sin^3 \theta = 3 \sin \theta - \sin 3\theta)$$

$$= 15 \sin \omega t - 5 (3 \sin \omega t - \sin 3\omega t)$$

$$= 15 \sin \omega t - 15 \sin \omega t + 5 \sin 3\omega t = 5 \sin 3\omega t$$

$\Rightarrow$ motion is SHM with angular frequency 3ω

$$\text{So } a_{\max} = 45\omega^2$$

10. **Ans (D)**

$$V_{A_1} = \frac{-kp}{\sqrt{2}r^2}$$

$$V_{A_2} = \frac{kp}{\sqrt{2}r^2}$$

$$\therefore V_{\text{net}} = 0$$

12. **Ans (C)**

Apply kirchoff law

for larger loop

$$iR_1 + iR_3 = v_1 + v_2$$

$$i = \frac{v_1 + v_2}{R_1 + R_3} \dots\dots\dots (1)$$

and for smaller loop

$$i = \frac{v_1}{R_1} \dots\dots\dots (2)$$

$$\text{solving } \frac{v_1}{v_2} = \frac{R_1}{R_3}$$

13. **Ans (B)**

$$\frac{V}{2R} = \frac{V}{R} e^{-t/\tau}$$

$$\frac{t}{\tau} = \ln 2$$

$$t = \tau \ln 2$$

15. **Ans (C)**

The volume of the cubic domain is 10^{-12}cc

its mass is volume density = 8 g/cm^3

It is given that Avagadro number (6×10^{23}) of iron atoms have a mass of 56g. Hence, the number of atoms in the domain is

$$N = \frac{8 \times 10^{-12} \times 6 \times 10^{23}}{56} = 0.82 \times 10^{11}$$

The maximum possible dipole moment m_{max} is achieved for the (unrealistic) case when all the atomic moments are perfectly aligned. Thus.

$$m_{\text{max}} = (8.2 \times 10^{10}) (9.1 \times 10^{-23})$$

$$= 7.8 \times 10^{-12}$$

$$M_{\text{max}} = m_{\text{max}} / \text{Domain volume}$$

16. **Ans (C)**

$$\cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{X_c^2 + R^2}}$$

$$= \frac{300}{500} = 0.6$$

$$P = V_{\text{rms}} i_{\text{rms}} \cos \phi$$

$$= \frac{V_{\text{rms}}^2}{Z} \cos \phi$$

$$= \frac{(200)^2}{500} \times 0.6$$

$$= 4.8 \text{ W}$$

18. **Ans (B)**

$$\lambda = \frac{12400}{E \text{ (eV)}}, \lambda = \frac{12400}{2} = 6100 \text{ \AA}$$

19. **Ans (A)**

The action of transistors is controlled by motions of charge carrier.

20. **Ans (D)**

Carrier frequency should be larger than modulating signal frequency

SECTION-II

1. **Ans (9.60)**

$$\text{F.B.D of pulley : } 3g - T = 3a$$

$$T - 2g = 2a$$

$$T = 2.4 \text{ g}$$

$$T' = 2T = 2.4g$$

Taking torque about A, $\tau_A = 0$

$$4.8g \times L = M' g \times \frac{1}{2}L \Rightarrow M' = 9.6$$

2. **Ans (23.60)**

By conservation of mechanical energy

$$\frac{1}{2} I \omega^2 = \left(mg \frac{\ell}{2} + mg \frac{\ell}{2} \right)$$

$$\frac{1}{2} \times \left(\frac{m\ell^2}{3} + m \left(\frac{\ell}{2} \right)^2 \right) \omega^2 = mg\ell$$

$$\frac{1}{2} \times \left(\frac{7m\ell^2}{12} \right) \omega^2 = mg\ell$$

$$\omega^2 = \frac{24g}{7\ell} = \frac{24 \times 10}{7 \times 3} = \frac{80}{7} \dots (1)$$

By conservation of angular momentum about the hinge

$$mv \cos 60^\circ \times \frac{\ell}{2} = \left[\frac{m\ell^2}{3} + m \left(\frac{\ell}{2} \right)^2 \right] \times \omega$$

$$mv \frac{\ell}{4} = \frac{7m\ell^2}{12} \times \omega$$

$$\omega = \frac{3v}{4\ell} = \frac{3 \times v}{4 \times 3} \dots (2)$$

by (1) and (2) $v = 4\sqrt{35} \text{ m/s}$

3. **Ans (1.50)**

$$\frac{h_1}{h_2} = \frac{T_1}{T_2} \cdot \frac{\rho_2}{\rho_1} = \frac{50}{25} \cdot \frac{0.3}{0.4} = \frac{3}{2}$$

4. **Ans (300.00)**

$$\text{Time constant} = \frac{L}{R_{eq}}$$

$$100 \times 10^{-6} = 40 \times 10^{-3} / R_{eq}$$

$$\left[\frac{600R}{600 + R} \right] + 200 = 400$$

$$R = 300 \Omega$$

5. **Ans (11.09 to 11.10)**

$$\text{For damped oscillation } \tau = \frac{2L}{R}$$

$$\text{half time} = \tau \times \ln 2$$

$$= \frac{2 \times 12 \times 10^{-3}}{1.5} \times 0.693$$

$$= 0.0111 = 11.1 \text{ ms}$$

6. **Ans (7.00)**

$$E_{ka} = 13.6 (z-1)^2 \left(1 - \frac{1}{4} \right) = 13.6 (z-1)^2 \times \frac{3}{4}$$

For P, $z = 10$

For Q, $z = 18$

So number of elements between P & Q = 7

7. **Ans (34.80)**

$$M_\infty = \frac{f_0}{f_e} \text{ and tube length} = f_0 + f_e$$

$$\frac{f_0}{f_e} = 5, f_0 + f_e = 36 \Rightarrow 6f_e = 36$$

$$f_e = 6 \text{ cm} \dots (1)$$

$$f_0 = 30 \text{ cm} \dots (2)$$

tube length for image at D

$$L = f_0 + \frac{Df_e}{D + f_e} = 30 + 4.8$$

$$= 34.8 \text{ cm}$$

8. **Ans (3.75)**

$$2 \mu t \cos r = n\lambda$$

$$2 \times \frac{4}{3} \times t \cos 60 = 1 \times 5 \times 10^{-7}$$

$$t = 3.75 \times 10^{-7} \text{ m}$$

9. **Ans (4.50)**

Most of the light is diffracted between the two first order minima.

$$b \sin \theta = \pm \lambda$$

$$\sin \theta = \pm \frac{\lambda}{b} = \pm 2.25 \times 10^{-3}$$

$$\text{So angular divergence} = 4.5 \times 10^{-3} \text{ rad}$$

10. **Ans (0.10)**

$$VSD = (0.4)^\circ$$

$$20 \times MSD = 25 \text{ VSD}$$

$$VSD = \frac{4}{5} MSD$$

$$L.C. = MSD - VSD$$

$$= \frac{5}{4} VSD - VSD = \frac{VSD}{4} = 0.1$$

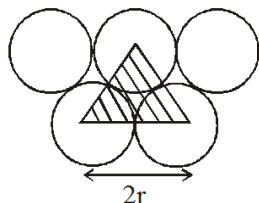
PART-2 : CHEMISTRY

SECTION-I

1. **Ans (B)**

If backward reaction is endothermic means forward reaction is exothermic hence on addition of Cl_2 reaction will go forward & temp. will increase.

2. **Ans (B)**



Effective area occupied by each NH_3 molecule

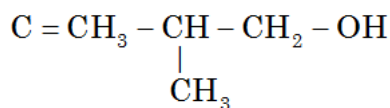
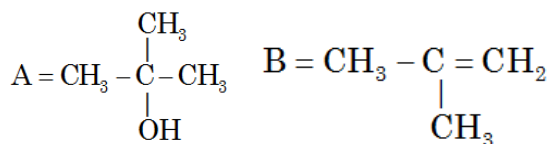
$$= \frac{\sqrt{3}}{4} \times (2r)^2 \times 2$$

$$= 2\sqrt{3} r^2 \text{ nm}^2$$

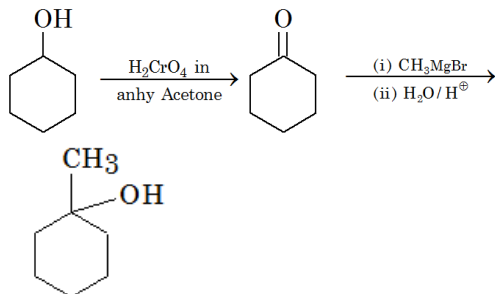
Number of ammonia molecule required

$$= \frac{20 \times 10^3}{2\sqrt{3} \times (r \times 10^{-9})^2} = \frac{10^{22}}{\sqrt{3} r^2}$$

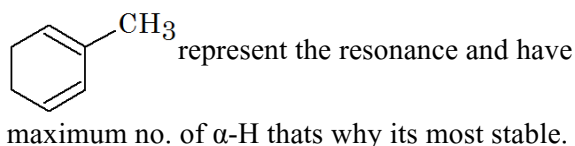
3. **Ans (C)**



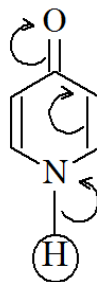
4. **Ans (D)**



5. **Ans (D)**

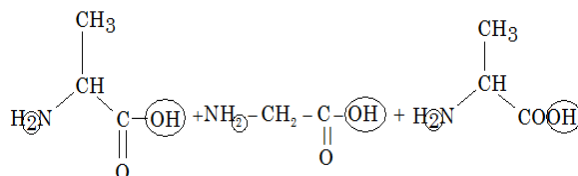
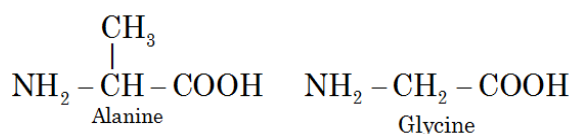


6. **Ans (C)**

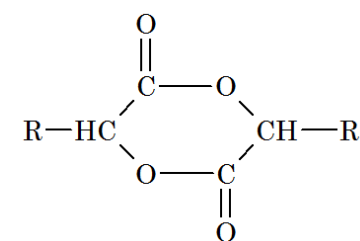
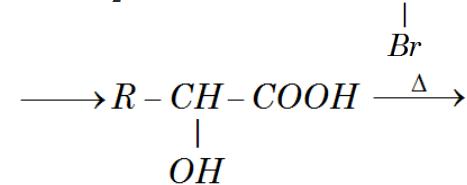
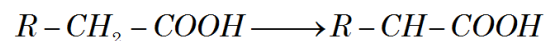


this circle acidic H migrate w.r.t π -bond & shows tautomerism.

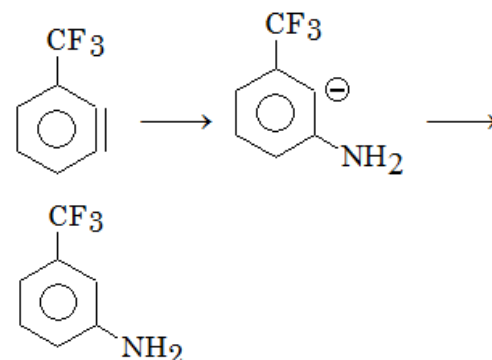
8. **Ans (C)**



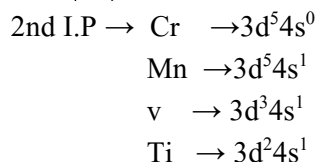
9. **Ans (C)**



11. **Ans (D)**



17. Ans (B)



SECTION-II

1. Ans (16.00)

$$\frac{r_A}{r_{SO_2}} = \frac{\eta_A}{\eta_{SO_2}} = \frac{\eta_A \text{ is container}}{\eta_{SO_2} \text{ is container}} \sqrt{\frac{M_{SO_2}}{M_A}}$$

$$\frac{1}{1} = \frac{1}{2} \sqrt{\frac{64}{M_A}}$$

$$4 = \frac{64}{M_A}, M_A = \frac{64}{4} = 16$$

2. Ans (16.00)

$$\Delta x \cdot \Delta p = \frac{h}{4\pi}$$

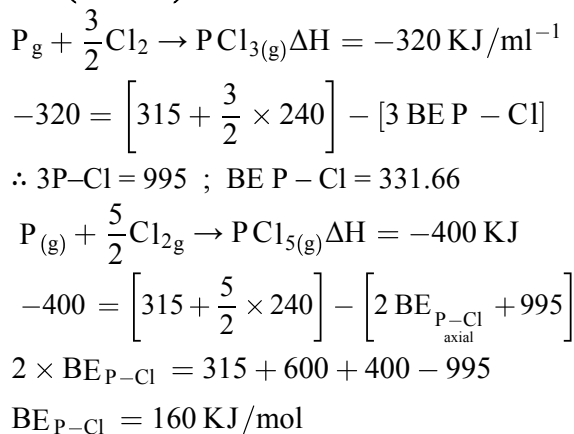
$$\Delta x = \frac{1}{2} \lambda = \frac{1}{2} \cdot \frac{h}{p}$$

$$\frac{1}{2} \cdot \frac{h}{p} \cdot \Delta p = \frac{h}{4\pi} \quad \% \text{ uncertainty}$$

$$\frac{\Delta p}{p} = \frac{1}{2\pi} = \frac{1}{2\pi} \times 100 = 15.9\%$$

$$\frac{m\Delta V}{mV} = \frac{1}{2\pi} \therefore \frac{\Delta V}{V} = \frac{1}{2\pi}$$

3. Ans (160.00)



4. Ans (400.00)

$$f_1 = e^{-\frac{E_a}{R \times 200}}$$

$$f_2 = e^{-\frac{E_a}{R \times 400}}$$

$$\frac{f_2}{f_1} = e^{1/2}$$

$$E_a = 200R$$

$$= 400 \text{ cal}$$

7. Ans (11.00)

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{n} \log Q$$

$$0.3 = E_{\text{cell}}^0 - \frac{0.06}{2} \log 10$$

$$E_{\text{cell}}^0 = 0.33 \text{ volt}$$

At equilibrium

$$\Delta G = 0 \quad E_{\text{cell}} = 0$$

$$Q = K$$

$$0 = 0.33 - 0.03 \log K$$

$$K = 10^{11}$$

8. Ans (1.00)

PH of salt of weak acid and weak base. remains independent of concentration.
Thus $\frac{x_1}{x_2} = 1$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

From the given data

$$f_1 = a_1 x_1 + a_2 x_2 + a_3 x_3$$

$$f_2 = b_1 x_1 + b_2 x_2 + b_3 x_3$$

$$f_3 = c_1 x_1 + c_2 x_2 + c_3 x_3$$

Since $f_1 = f_2 = f_3 = 0$, so this becomes in form of homogeneous linear equation

$$\therefore a_1 x_1 + a_2 x_2 + a_3 x_3 = 0$$

$$b_1 x_1 + b_2 x_2 + b_3 x_3 = 0$$

$$c_1 x_1 + c_2 x_2 + c_3 x_3 = 0$$

$$\text{so, } \Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

2. Ans (A)

Correct result is $(\sim p \vee \sim q) \Rightarrow (r \wedge s)$
So, $\sim(p \wedge q) \Rightarrow (r \wedge s)$.

3. **Ans (B)**

Prob that student is passed in mathematics

$$P(M) = 1/4$$

Prob that students is passed in physics

$$P(P) = 1/5$$

Prob. that student is passed in both the subjects

$$P(M \cap P) = 3/20$$

$$\text{required probability} = \frac{P(M \cap \bar{P})}{P(\bar{P})}$$

$$= \frac{P(M) - P(M \cap P)}{1 - P(P)}$$

$$= \frac{\frac{1}{4} - \frac{3}{20}}{1 - \frac{1}{5}} = \frac{1/10}{4/5} = \frac{1}{8}$$

4. **Ans (B)**

Since $|z_1| = 12$

$$|z_2 - \sqrt{3} - i| = 5$$

$$= |z_2 - (\sqrt{3} + i)| = 5 \geq |z_2| - |\sqrt{3} + i|$$

$$5 \geq |z_2| - 2$$

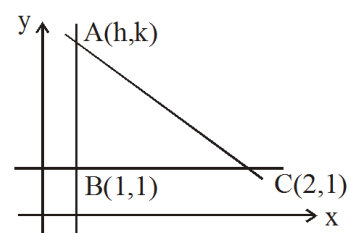
$$7 \geq |z_2|$$

$$\text{Now } ||z_1| - |z_2|| \leq |z_1 - z_2| \leq |z_1| + |z_2|$$

$$|12 - 7| \leq |z_1 - z_2| \leq 12 + 7$$

$$5 \leq |z_1 - z_2| \leq 19$$

5. **Ans (C)**



$$\therefore \frac{1}{2} \times (1) \times (k-1) = \pm 2$$

$$k-1 = \pm 4$$

$$\begin{array}{l} \text{ } \\ \text{ } \end{array}$$

$$k = 1 + 4 \quad k = 1 - 4$$

$$k = 5 \quad k = -3$$

$$\therefore \text{values of } h^2 + k^2 = 1^2 + 5^2 = 26$$

or

$$h^2 + k^2 = 1^2 + 9 = 10$$

6. **Ans (B)**

$$3x + 4y + z - 3 = 0 \quad \dots(1)$$

$$3x + 4y + z - \frac{5}{2} = 0 \quad \dots(2)$$

Let equation of plane is

$$3x + 4y + z + k = 0 \quad \dots(3)$$

Given that

$$\Rightarrow \frac{\left| \frac{k+3}{\sqrt{26}} \right|}{\left| \frac{k+\frac{5}{2}}{\sqrt{26}} \right|} = \frac{2}{3} \Rightarrow 3|k+3| = 2 \left| k + \frac{5}{2} \right|$$

$$\Rightarrow 3k + 9 = -2k - 5 \quad (\text{Taking -ve sign})$$

$$\Rightarrow k = \frac{-14}{5}$$

Equation required plane is

$$3x + 4y + z - \frac{14}{5} = 0$$

$$15x + 20y + 5z - 14 = 0$$

7. **Ans (C)**

$$\text{Let } p(x) = x^2 - 4x + 3$$

$$\frac{dp}{dx} = 2x - 4 = 0 \Rightarrow x = 2$$

$$p(2) = 3, p(5) = 8, p(-5) = 48$$

$$(f(x))_{\max} = f(-5) = \log_{\sqrt{48}} (25 + 20 + 3)$$

$$= \log_{\sqrt{48}} 48 = 2$$

8. **Ans (D)**

$$\tan^{-1} \frac{3}{5} + \tan^{-1} \frac{3}{29} + \tan^{-1} \frac{3}{71} + \tan^{-1} \frac{3}{131} + \dots n \text{ terms}$$

$$= \tan^{-1} \left(\frac{4-1}{1+4 \times 1} \right) + \tan^{-1} \left(\frac{7-4}{1+7 \times 4} \right) +$$

$$\tan^{-1} \left(\frac{10-7}{1+10 \times 7} \right) + \dots \tan^{-1} \left(\frac{3n+1-(3n-2)}{1+(3n+1)(3n-2)} \right)$$

$$= \tan^{-1} 4 - \tan^{-1} 1 + \tan^{-1} 7 - \tan^{-1} 4 + \tan^{-1} 10$$

$$- \tan^{-1} 7 + \dots \tan^{-1} (3n+1) - \tan^{-1} (3n-2)$$

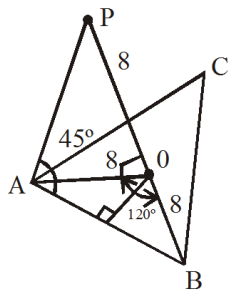
$$= \tan^{-1} (3n+1) - \tan^{-1} 1 = \tan^{-1} \left(\frac{3n+1-1}{1+(3n+1)} \right)$$

$$= \tan^{-1} \left(\frac{3n}{2+3n} \right)$$

9. **Ans (B)**

$$A = \int_0^1 (2x - x) dx + \int_1^4 (2\sqrt{x} - x) dx = \frac{7}{3}$$

10. Ans (C)



$$\begin{aligned}\text{Area of } \triangle ABC &= 3 \times \text{Area of triangle OAB} \\ &= 3 \times \frac{1}{2} \times 8 \times 8 \times \sin 120^\circ \\ &= 3 \times 4 \times 8 \times \frac{\sqrt{3}}{2} = 48\sqrt{3}\end{aligned}$$

11. Ans (B)

$$\begin{aligned}r &= a_1 + \lambda b_1 \text{ and } r = a_2 + \lambda b_2 \\ \text{then S. D.} &= \left| \frac{(a_2 - a_1) \cdot (b_1 \times b_2)}{|b_1 \times b_2|} \right| \\ \Rightarrow \ell &= \frac{6}{\sqrt{5}} \\ \text{then } [\ell] &= 2\end{aligned}$$

12. Ans (C)

$$\begin{aligned}|\text{adj} A| &= |A|^{n-1} \\ \Rightarrow |\text{adj}(\text{adj} A)| &= |\text{adj} A|^{n-1} = |A|^{(n-1)^2} \\ \Rightarrow (n-1)^2 &= P^2 \Rightarrow n-1 = P \\ \Rightarrow n &= P+1\end{aligned}$$

13. Ans (C)

$$\begin{aligned}a + 8d &= b \quad \dots (1) \\ a + 12d &= br \quad \dots (2) \\ a + 14d &= br^2 \quad \dots (3) \\ \Rightarrow (2) - (1) &\Rightarrow 4d = b(r-1) \\ (3) - (2) &\Rightarrow 2d = br(r-1) \\ \Rightarrow 2 &= \frac{1}{r} \Rightarrow r = \frac{1}{2} \\ \text{also } \frac{b}{1-r} &= 80 \Rightarrow b = 40 \quad d = -5 \Rightarrow a = 80\end{aligned}$$

14. Ans (D)

$$\begin{aligned}\text{Let point of intersection be } (h, k) \text{ then equation of PQ is } hx + 2ky &= 4, \text{ it touches } x^2 + y^2 = 1 \\ \Rightarrow \frac{16}{4k^2} &= 1 + \frac{h^2}{4k^2} (\because c^2 = a^2(1+m^2)) \\ \Rightarrow \frac{x^2}{16} + \frac{y^2}{4} &= 1 \Rightarrow e = \frac{\sqrt{3}}{2}\end{aligned}$$

15. Ans (B)

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$\text{Condition of tangency } c = \pm \sqrt{a^2 m^2 - b^2}$$

$$2 = \pm \sqrt{9m^2 - 4}$$

$$4 = 9m^2 - 4$$

$$9m^2 = 8$$

$$m = \pm \frac{2\sqrt{2}}{3}$$

16. Ans (D)

$$y^2 = x^3$$

$$\frac{dy}{dx} = \frac{3x^2}{2y}$$

$$\text{slope of tangent at } (\alpha^2, \alpha^3) = \text{slope of normal}$$

$$\text{at } (\beta^2, \beta^3)$$

$$\frac{3\alpha^4}{2\alpha^3} = \frac{-1}{\frac{3\beta^4}{2\beta^3}}$$

$$\frac{3}{2}\alpha = \frac{-2}{3\beta}$$

$$\alpha\beta = \frac{-4}{9}$$

17. Ans (C)

$$(1+a)^x = {}^x C_0 + {}^x C_1 a + {}^x C_2 a^2 + \dots + {}^x C_r a^r + \dots + {}^x C_x a^x \dots (1)$$

$$(1+a)^y = {}^y C_0 + {}^y C_1 a + {}^y C_2 a^2 + \dots + {}^y C_r a^r + \dots + {}^y C_y a^y \dots (2)$$

multiplying eq.(1) and (2) and equating the

coefficient of a^r both sides,

$${}^x C_r {}^y C_0 + {}^x C_{r-1} {}^y C_1 + {}^x C_{r-2} {}^y C_2 + \dots + {}^x C_0 {}^y C_r$$

$$= \text{coefficient of } a^r \text{ in } (1+a)^{x+y} = {}^{x+y} C_r$$

18. Ans (D)

$$np = 4, npq = 2$$

$$q = \frac{1}{2}, p = 1 - q = \frac{1}{2}$$

$$\therefore n = 8$$

$$P(x=1) = {}^8 C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^7 = 8 \times \frac{1}{2^8} = \frac{1}{2^5} = \frac{1}{32}$$

19. Ans (B)

Let the number is abc

$$\text{such that } b = \frac{a+c}{2}$$

$$2b = a + c$$

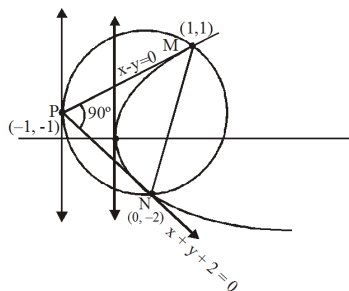
$$\text{even} = a + c$$

a, c should be both even or both odd.

$$\text{No. of ways} \Rightarrow 5.5 + 4.5$$

$$= 25 + 20 = 45$$

20. Ans (A)



MN will be focal chord of parabola. A circle with MN as diameter will touch directrix at P(-1, -1).

Equation of circle with MN as diameter.

$$(x-1)x + (y-1)(y+2) = 0$$

$$\Rightarrow x^2 + y^2 - x + y - 2 = 0$$

Equation of tangent at P(-1, -1) will be

$$x(-1) + y(-1) - \frac{(x-1)}{2} + \frac{(y-1)}{2} - 2 = 0$$

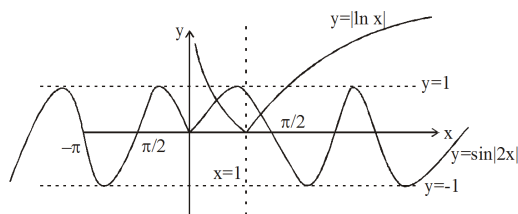
$$\Rightarrow -3x - y - 4 = 0$$

$$\Rightarrow 3x + y + 4 = 0$$

SECTION-II

1. Ans (2.00)

From the graphs of $y = f(|x|)$ and $y = |g(x)|$



so total no. of solutions will be two in $x \in (0, \pi)$

2. Ans (4.25)

$$\frac{dy}{dx} + \frac{3y}{x} = 1$$

$$\text{I.F.} = e^{\int \frac{3dx}{x}} = e^{3 \log x} = x^3$$

solution of differential equation

$$y \cdot x^3 = \int x^3 dx \Rightarrow yx^3 = \frac{x^4}{4} + c$$

put $x = 2, y = 1$, we get $c = 4$

$$yx^3 = \frac{x^4}{4} + 4$$

$$\text{Put } x = 1, y(1) = \frac{1}{4} + 4 = \frac{17}{4}$$

3. Ans (1.00)

$$A_n = \int_0^{\pi/2} \frac{\sin(2n-1)x}{\sin x} dx$$

$$A_{n+1} - A_n = \int_0^{\pi/2} \frac{(\sin(2n+1)x - \sin(2n-1)x)}{\sin x} dx$$

$$= \int_0^{\pi/2} 2 \cos 2nx dx = 0$$

$$\Rightarrow A_{n+1} = A_n$$

$$\Rightarrow A_1 = A_2 = A_3 = \dots = A_{100} = \dots = A_{1000}$$

4. Ans (7.50)

$$\lim_{x \rightarrow 0} \frac{1}{2} \left(\frac{2 \cos 5x \cos 2x - 2 \cos 7x \cos 10x}{\ln^2(1 + \sin 2x)} \right)$$

$$\lim_{x \rightarrow 0} \frac{1}{2} \left(\frac{\cos 7x + \cos 3x - \cos 17x - \cos 3x}{\sin^2 2x} \right)$$

$$\lim_{x \rightarrow 0} \frac{1}{2} \left(\frac{\cos 7x - \cos 17x}{\sin^2 2x} \right)$$

$$\lim_{x \rightarrow 0} \frac{1}{2} \frac{2 \sin 12x \sin 5x}{\sin^2 2x}$$

$$= \frac{12.5}{2.2} = 15$$

5. Ans (1.50)

$$\vec{x} \times \vec{z} + \vec{y} = \vec{0} \text{ (Given)}$$

Taking cross product with $\vec{x}$ both sides,

$$\vec{x} \times (\vec{x} \times \vec{z} + \vec{y}) = \vec{x} \times \vec{0}$$

$$\vec{x} \times (\vec{x} \times \vec{z}) + \vec{x} \times \vec{y} = \vec{0}$$

$$(\vec{x} \cdot \vec{z}) \vec{x} - (\vec{x} \cdot \vec{x}) \vec{z} + \vec{x} \times \vec{y} = \vec{0}$$

$$\Rightarrow 2\vec{z} = \vec{x} \times \vec{y}$$

$$\text{Now } \vec{x} \times \vec{y} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = -\hat{i} - \hat{j} + 2\hat{k}$$

$$\therefore 2\vec{z} = -\hat{i} - \hat{j} + 2\hat{k}$$

$$2|\vec{z}| = \sqrt{6}$$

$$|\vec{z}| = \sqrt{\frac{3}{2}} \Rightarrow |\vec{z}|^2 = \frac{3}{2}$$

6. **Ans (5.33)**

$$\because f^{-1}(2) = 3, f^{-1}(-3) = 6$$

$$\Rightarrow f(3) = 2, f(6) = -3$$

$$3a + b = 2$$

$$+6a + b = -3$$

Solving eq. (1) & (2)

$$a = -\frac{5}{3}, b = 7$$

$$\therefore a + b = -\frac{5}{3} + 7 = \frac{-5 + 21}{3} = \frac{16}{3}$$

7. **Ans (2.50)**

Let E_1 be the event that one of first n urns is chosen

E_2 be the event that $(n + 1)^{\text{th}}$ urn is chosen

A be the event that two balls drawn are black then

$$P(E_1) = \frac{n}{n+1}, P(E_2) = \frac{1}{n+1}, P\left(\frac{A}{E_1}\right) = \frac{{}^6C_2}{{}^{10}C_2} = \frac{1}{3}$$

$$P\left(\frac{A}{E_2}\right) = \frac{{}^5C_2}{{}^{10}C_2} = \frac{2}{9}$$

$$\text{Now } P\left(\frac{E_2}{A}\right) = \frac{\frac{1}{n+1} \cdot \frac{2}{9}}{\frac{n}{n+1} \cdot \frac{1}{3} + \frac{1}{n+1} \cdot \frac{2}{9}}$$

$$\frac{1}{16} = \frac{2}{3n+2}$$

$$3n + 2 = 32 \Rightarrow n = 10$$

8. **Ans (0.00)**

Put $x = 2$

$$\Rightarrow 2f(2) + 2f\left(\frac{1}{2}\right) - 2f(1) = 4$$

$$\Rightarrow f(2) + f\left(\frac{1}{2}\right) - f(1) = 2 \dots (1)$$

Put $x = 1$

$$\Rightarrow f(1) = -1 \dots (2)$$

$$\text{Put } x = \frac{1}{2}$$

$$2f\left(\frac{1}{2}\right) + \frac{1}{2}f(2) - 2f(1) = \frac{5}{2} \dots (3)$$

solving (1), (2), (3)

$$f(2) = 1, f\left(\frac{1}{2}\right) = 0$$

9. **Ans (1.00)**

Difference of roots in both equation is same

i.e., $\cos 2\theta$

$$\Rightarrow 16 - 8 = 4b^2 - 4b$$

$$\Rightarrow 2 = b^2 - b$$

$$\Rightarrow b^2 - b - 2 = 0$$

$$\Rightarrow (b - 2)(b + 1) = 0 \Rightarrow b = 2, -1$$

10. **Ans (0.00)**

$$I = \int_0^1 \sin 8\pi x \cdot g(x) dx$$

$$= \int_0^{1/2} \sin 8\pi x \cdot g(x) dx + \int_{1/2}^1 \sin 8\pi x \cdot g(x) dx$$

$$= I_1 + I_2$$

$$I_2 = \int_{1/2}^1 \sin 8\pi x \cdot g(x) dx$$

$$\text{Put } x = \frac{1}{2} + t \quad dx = dt$$

$$I_2 = \int_0^{1/2} \sin 8\pi \left(t + \frac{1}{2}\right) \cdot g\left(t + \frac{1}{2}\right) dt$$

$$= \int_0^{1/2} \sin 8\pi t (2 - g(t)) dt$$

$$= 2 \int_0^{1/2} \sin 8\pi t \cdot dx - \int_0^{1/2} (\sin 8\pi t) g(t) dt$$

$$\Rightarrow I = 2 \int_0^{1/2} \sin 8\pi t dt$$

$$= 2 \left[-\frac{\cos 8\pi t}{8\pi} \right]_0^{1/2} = 0$$